

**SEMESTER S5**  
**CAD FOR VLSI DESIGN**

|                                            |                 |                    |                |
|--------------------------------------------|-----------------|--------------------|----------------|
| <b>Course Code</b>                         | <b>PEEVT524</b> | <b>CIE Marks</b>   | 40             |
| <b>Teaching Hours/Week<br/>(L: T:P: R)</b> | 3:0:0:0         | <b>ESE Marks</b>   | 60             |
| <b>Credits</b>                             | 3               | <b>Exam Hours</b>  | 2 Hrs. 30 Min. |
| <b>Prerequisites (if any)</b>              | None            | <b>Course Type</b> | Theory         |

**Course Objectives:**

1. Understand the different stages of design flow; the basic data structures and algorithms used in each stage
2. Understand data structures and algorithms used in recent CAD tools
3. Identify suitable data structures and propose new algorithms for CAD applications and develop new CAD tools

**SYLLABUS**

| <b>Module No.</b> | <b>Syllabus Description</b>                                                                                                                                                                                                                                                                                                                           | <b>Contact Hours</b> |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <b>1</b>          | Graph Terminology: Basic graph theory terminology, Data structures for representation of Graphs Search Algorithms: Breadth First Search, Depth First Search, Topological Sort Shortest Path Algorithms: Dijkstra's Shortest-Path Algorithm for single pair shortest path, Floyd Warshall Algorithm for all pair shortest path                         | 9                    |
| <b>2</b>          | VLSI Design Flow, VLSI Design Styles<br>Partitioning: Levels of Partitioning, Parameters for Partitioning, Classification of Partitioning Algorithms, Kernighan-Lin Algorithm, Fiduccia-Mattheyses Algorithm                                                                                                                                          | 9                    |
| <b>3</b>          | <b>Layout:</b> Layout Layers and Design Rules, Physical Design Optimizations<br>Compaction: Applications of Compaction, Informal Problem Formulation, Graph Theoretical Formulation, Maximum Distance Constraints, Longest Path algorithm for DAG, Liao-Wong Algorithm.<br><b>Placement:</b> Optimization Objectives, Wirelength Estimation, Weighted | 9                    |

|          |                                                                                                                                                                                                                                                                                                                                                                                                                              |          |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
|          | Wirelength, Maximum Cut Size, Wire Density<br>Placement Algorithms: Quadratic Placement                                                                                                                                                                                                                                                                                                                                      |          |
| <b>4</b> | <b>Floorplanning:</b> Optimization Objectives, Slicing Floorplan, Non-Slicing Floorplan. Floorplan Representations: Constraint Graph, Sequence Pair<br><b>Global Routing:</b> Terminology and Definitions, Optimization Goals, Representation of Routing Regions<br>Maze Routing Algorithms: Lee's Algorithm,<br>Detailed Routing: Horizontal and Vertical Constraint Graph. Channel Routing Algorithms: Left-Edge algorithm | <b>9</b> |

**Course Assessment Method**  
**(CIE: 40 marks, ESE: 60 marks)**

**Continuous Internal Evaluation Marks (CIE):**

| <b>Attendance</b> | <b>Assignment/<br/>Microproject</b> | <b>Internal<br/>Examination-1<br/>(Written)</b> | <b>Internal<br/>Examination- 2<br/>(Written )</b> | <b>Total</b> |
|-------------------|-------------------------------------|-------------------------------------------------|---------------------------------------------------|--------------|
| <b>5</b>          | <b>15</b>                           | <b>10</b>                                       | <b>10</b>                                         | <b>40</b>    |

**End Semester Examination Marks (ESE)**

*In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions*

| <b>Part A</b>                                                                                                                                                                                 | <b>Part B</b>                                                                                                                                                                                                                                                                                                | <b>Total</b> |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| <ul style="list-style-type: none"> <li>2 Questions from each module.</li> <li>Total of 8 Questions, each carrying 3 marks</li> </ul> <p style="text-align: center;"><b>(8x3 =24marks)</b></p> | <ul style="list-style-type: none"> <li>Each question carries 9 marks.</li> <li>Two questions will be given from each module, out of which 1 question should be answered.</li> <li>Each question can have a maximum of 3 sub divisions.</li> </ul> <p style="text-align: center;"><b>(4x9 = 36 marks)</b></p> | <b>60</b>    |

### Course Outcomes (COs)

At the end of the course students should be able to:

| Course Outcome |                                                                                       | Bloom's Knowledge Level (KL) |
|----------------|---------------------------------------------------------------------------------------|------------------------------|
| <b>CO1</b>     | Apply Search Algorithms and Shortest Path Algorithms to find various graph solutions. | <b>K3</b>                    |
| <b>CO2</b>     | Apply partitioning algorithms on graphs representing netlist.                         | <b>K3</b>                    |
| <b>CO3</b>     | Apply different algorithms for layout compaction and placement                        | <b>K3</b>                    |
| <b>CO4</b>     | Utilise different algorithms to solve floorplan and routing problems.                 | <b>K3</b>                    |

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

### CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

|            | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
| <b>CO1</b> | 3   | 3   | 2   |     |     |     |     |     |     | 1    |      |      |
| <b>CO2</b> | 3   | 3   | 2   |     |     |     |     |     |     | 1    |      |      |
| <b>CO3</b> | 3   | 3   | 2   |     |     |     |     |     |     | 1    |      |      |
| <b>CO4</b> | 3   | 3   | 2   |     |     |     |     |     |     | 1    |      |      |
| <b>CO5</b> |     |     |     |     |     |     |     |     |     |      |      |      |

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

| Text Books |                                                                 |                                                      |                            |                  |
|------------|-----------------------------------------------------------------|------------------------------------------------------|----------------------------|------------------|
| Sl. No     | Title of the Book                                               | Name of the Author/s                                 | Name of the Publisher      | Edition and Year |
| 1          | “Algorithms for VLSI Design Automation”,                        | Gerez, Sabih H.                                      | John Wiley & Sons          | 2006.            |
| 2          | Algorithms for VLSI Physical Design Automation                  | Sherwani, Naveed A                                   | Kluwer Academic Publishers | 1999             |
| 3          | VLSI Physical Design: From Graph Partitioning to Timing Closure | Jin Hu, Jens Lienig, Igor L. Markov, Andrew B. Kahng | Springer                   | 2011             |

| Reference Books |                                                      |                                                               |                       |                   |
|-----------------|------------------------------------------------------|---------------------------------------------------------------|-----------------------|-------------------|
| Sl. No          | Title of the Book                                    | Name of the Author/s                                          | Name of the Publisher | Edition and Year  |
| 1               | VLSI Physical Design Automation: Theory and Practice | Sadiq M. Sait and H. Yousse                                   | World Scientific      | 1999              |
| 2               | Introduction to Algorithms.                          | Cormen, Thomas H., Charles E. Leiserson, and Ronald L. Rivest | The MIT Press         | 3rd edition, 2009 |

| Video Links (NPTEL, SWAYAM...) |                                                                                                       |
|--------------------------------|-------------------------------------------------------------------------------------------------------|
| Module No.                     | Link ID                                                                                               |
| 1                              | <a href="https://youtu.be/hk5rQs7TQ7E?feature=shared">https://youtu.be/hk5rQs7TQ7E?feature=shared</a> |
| 2                              | <a href="https://youtu.be/O9guSe5_tG0">https://youtu.be/O9guSe5_tG0</a>                               |
| 3                              | <a href="https://youtu.be/F44WOxhbtV0">https://youtu.be/F44WOxhbtV0</a>                               |
| 4                              | <a href="https://youtu.be/TYy2o8Qy4TY">https://youtu.be/TYy2o8Qy4TY</a>                               |